



Fig. 4: (a) TVWS base station, (b) TVWS customer premises equipment (CPE) and (c) Pruthvi chip (Image courtesy: Saankhya Labs)

beginning to believe in the white-space approach to connecting areas where traditional broadband is not feasible.

White-space refers to the gaps left by television networks. Usually, television networks leave gaps between channels for buffering purposes, leading to some unused frequencies in the wireless spectrum. This space is similar to what is used for 4G, and can therefore be effectively used to deliver broadband Internet. White-space broadband is somewhat like Wi-Fi, just that it can travel a much longer distance of approximately 10 kilometres.

"The unused TV spectrum in the ultra-high-frequency (UHF) band, called white-spaces, has very good propagation characteristics, which makes it attractive for long-range communications at low power. These properties make this spectrum particularly attractive to provide affordable wireless broadband connectivity to large rural and remote areas, which are not served today.

"A single television white-space base station can reach households that are as far away as 5km to 10km, depending on antenna height. Bridging the broadband digital di-

vide between rural and urban areas is a must have for the success of Digital India, and television white-space modems play a vital role here," explains Vishwa Kayargadde, co-founder and chief scientist, Saankhya Labs.

You need special modems to use television white-space, and Saankhya Labs is a forerunner in this industry. Pruthvi is a semiconductor chipset developed in India by Saankhya Labs. It is a unique class of chip called software-defined radio (SDR), which allows the same hardware to be used for different types of wireless communication by loading appropriate software.

Pruthvi contains highly-specialised signal-processing central processing units (CPUs) and hardware engines designed by Saankhya from the ground up. These CPUs are very powerful and can perform tens of giga operations per second in a small silicon area while consuming low power, which allows broadband wireless communication to be performed by embedded software running on this chip.

Pruthvi SDR has been used in various applications, including UHD set-top boxes, television

dongles, video-distribution/surveillance drones, satellite receivers and Saankhya's Meghdoot base stations and user modems that provide rural broadband using television white-space.

Meghdoot is a complete VHF/UHF base station modem consisting of radio-front-end and baseband processing, powered by Pruthvi.

"Unlike other solutions powered by field-programmable gate arrays (FPGAs) and general-purpose digital signal processors (DSPs), Saankhya's solution is the first television white-space modem to be powered by a communication system on chip (SoC). Also, being a designed-in-India and made-in-India solution, we have several advantages besides easy access to the local market," says Kayargadde.

The company is currently performing pilot tests of their solution at various locations in India, in collaboration with various IITs. Unlike the USA, the UK, Singapore and several other nations, India is yet to release television white-space spectrum for unlicensed access. It is expected to be released in the near future. Until that spectrum is made available for unlicensed access similar to the Wi-Fi spectrum, large-scale proliferation of television white-space devices is unlikely to happen.

However, television white-space is a very promising technology for connecting rural areas. It has been tested successfully in several university campuses and districts across the world, after Microsoft, BBC, BT and Nokia launched a consortium to support the project in 2011.

Microsoft 4Africa is also betting big on white-space technology to connect interior regions of Africa, and so is Google.

Kayargadde believes that it will do wonders for India, too, even better than balloons or drones. He says, "Television white-space modem technology can be used to provide